

measure of the camera's horizontal resolution. Cameras with a high TVL resolution can capture highly detailed images. Most FPV cameras work on analogue video, and that's why the measurement is in TVL, rather than actual pixel counts common with digital video. Common TVL resolutions for FPV cameras are 20, 480, 600, 800 and 1000.

**Aspect ratio.** There are two common aspect ratios to choose from, 4:3 and 16:9. While selecting, keep an eye on the native aspect ratio of your FPV monitor/goggles. This is crucial because if you have a 4:3 camera but 16:9 display, the image appears debased, and if you have a 16:9 camera but 4:3 display, the image appears crushed.

**Video encoding.** PAL or NTSC? Actually, it's not a big issue as both PAL and NTSC encoding standards are supported by most FPV gears. PAL offers higher resolution, while NTSC allows higher frame rate. So if you want to have good-quality pictures, consider PAL.

**WDR.** The wide dynamic range (WDR) of an FPV camera dictates how well the camera can record details in very dark and very bright portions of the image. Nowadays most FPV cameras have wide dynamic range by default.

**OSD.** An on-screen display (OSD) device overlays text/data onto the camera footage. So it can display highly useful information such as the flight speed, altitude, distance and battery status on the FPV display screen.

**MIC.** Sound is a big part of the FPV flying experience, so if you have an FPV camera with built-in microphone you get many benefits. For example, you can hear the throttle response even when you are far away.

**IR filter/LEDs.** As with almost all other camera types, FPV cameras have an infrared (IR) filter built into them to block out any infrared light from reaching the camera sensor.



Fig. 2: An FPV camera module



Fig. 3: FPV transmitter and receiver modules



Fig. 4: FPV video monitor and goggles



Fig. 5: FPV on-screen display



Fig. 6: FPV camera with MIC and IR LEDs

However, there are some odd situations when you want to have an infrared-sensitive camera, especially at low light levels. So if you want to fly when it's darker, look for a

starlight FPV camera, which has a very low lux sensitivity and hence can work in very low lighting conditions. As an alternative, you can try a standard FPV camera fitted with some infrared light-emitting diodes (IRLEDs) on its front side.

There are some exceptional 3D FPV cameras, too. In principle, a 3D FPV camera is a perfect blend of two FPV cameras to make the illusion of an actual flight.

## Other things to consider

In the market, you can find several FPV drones with cameras and video transmitters already installed, and video displays built right into the radio-controllers. However, by selecting your own airborne and ground station components, you can put together a much higher performing and flexible drone rig that grows with you as you move on to bigger ones. Well, here is some other practical information about the supporting components of an FPV camera.

### Radio frequency and channel.

FPV transmitter frequency is the most important thing to consider. Nowadays most FPV equipment run on 5.8GHz, as it is allowed for FPV use in most parts of the globe. Also 5.8GHz antennae are much smaller, which makes 5.8GHz transmitters cheaper and easier to use. You can also use 2.4GHz band. However, since almost all radio-controlled (RC) equipment run on 2.4GHz, you will receive lots of interference in this band.

Today, it's common to find 32-channel (32CH) and even 40-channel (40CH) FPV transmitters. Having a multiple-channel FPV transmitter is useful when flying with other FPV pilots, as each pilot can fly on a different frequency and hence your FPV video will not interfere with theirs. Note that each transmitter has a frequency table that lists the channels, bands and corresponding frequencies.